UNICC AI Agent: A Multi-Agent AI Governance Evaluation System

Applied Technical Project

Final Report

by

Yizhuo Shi

NYU NetID: ys3843

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Abstract

This Applied Technical Project designed, implemented, and evaluated UNICC AI Agent, a local multi-agent AI governance evaluation system intended to support responsible review of proposed AI use cases. The project responded to a growing organizational need: as generative AI and agentic AI systems move from experimentation into operational workflows, institutions require a structured method for evaluating regulatory alignment, safety risk, assurance evidence, accountability, and deployment readiness before systems are approved for use. The solution uses a FastAPI-based backend, multiple specialized governance agents, an orchestrator that aggregates findings, and a dashboard-oriented user experience. The architecture evaluates AI use cases through Standards Compliance, Safety Dimensions, and Assurance & Accountability perspectives, then produces one of four decision outcomes: Approve, Approve with Conditions, Flag for Review, or Reject. The reference implementation uses Llama 3.2 through Ollama for local inference, which supports the project objective of avoiding external API dependence during governance review. The methodology combined requirements analysis, architecture design, prompt engineering, API implementation, output normalization, scenario-based testing, and risk review. The project demonstrated that a small, modular AI governance system can make AI review more consistent, explainable, and auditable than an ad hoc manual review process. The

project also showed important limits: model outputs still require human validation, local models vary in reasoning quality, standards evolve quickly, and the application remains a decision-support tool rather than an autonomous approval authority. Recommendations include improving test coverage, adding persistent storage and role-based access controls, strengthening evidence citation, introducing human-in-the-loop workflow states, and maintaining a regulatory knowledge base. The completed project provides a practical foundation for institutional AI governance simulation and future development into a more robust responsible AI review platform.

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Declaration

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Abbreviations and Glossary

Table 0-1. Abbreviations and glossary

Term	Meaning
AI	Artificial intelligence.
API	Application programming interface.
FastAPI	A Python framework used to implement backend services and interactive API documentation.
LLM	Large language model.
Ollama	A local model serving tool used to run Llama 3.2 in the reference implementation.
Orchestrator	The service that calls specialized agents, normalizes their outputs, aggregates risks and controls, and derives a final governance decision.
Pydantic	A Python validation library used to define request and response schemas.
SLM	Small language model, used here to refer to local open-weight models appropriate for constrained governance experiments.
UNICC	United Nations International Computing Centre, used as the applied project reference context.

1. Introduction

UNICC AI Agent is an applied prototype for evaluating proposed AI systems before they are deployed into sensitive organizational workflows. The project was developed as a governance-oriented software system rather than as a generic chatbot. Its purpose is to help an institution convert broad responsible AI principles into a repeatable review process that can be executed through a local backend, specialized evaluation agents, and a consolidated decision engine.

The reference repository identifies the system as a multi-agent AI governance evaluation system. The repository overview describes its scope across regulatory compliance, safety risks, and assurance and accountability. It also describes the architecture as a set of specialized agents coordinated by an orchestrator, with explainable outputs, cross-agent risk aggregation, a final decision engine, a React dashboard, a FastAPI backend, and local LLM support through Llama 3.2 via Ollama. These design choices shaped the project narrative and evaluation criteria in this final report.

The company and sponsor context for this report is UNICC / NYU, with Dr. Andres Fortino identified as the project sponsor. In an institutional technology environment, the project is best understood as a proof-of-value governance tool: it does not merely show that an LLM can answer safety questions, but demonstrates how a structured review

workflow could help decision makers screen AI use cases, document risks, and impose conditions before deployment. The project therefore sits at the intersection of enterprise architecture, responsible AI governance, and applied software engineering.

The project uses a modular architecture. A Standards Compliance Agent evaluates regulatory and standards alignment; a Safety Dimensions Agent evaluates harmfulness, bias and fairness, transparency, deception, privacy, and accountability; and an Assurance & Accountability Agent evaluates evidence quality, monitoring, ownership, escalation, and residual deployment risk. The orchestrator then combines those results and produces a final decision. This division of labor allows the system to model a committee-style review process while still producing a single consolidated output.

The intended users include AI governance teams, compliance analysts, technology managers, project sponsors, academic reviewers, and prototype evaluators. The system is most valuable before an AI application is deployed, when stakeholders need a fast but structured decision about whether a use case is ready, conditionally ready, questionable, or inappropriate. It is not a substitute for legal counsel, security review, privacy review, human subjects review, or senior governance approval. Instead, it is a decision-support layer that organizes review evidence and improves consistency across proposed AI use cases.

Fig. 1-1. UNICC AI Agent high-level architecture

Layer	Component	Purpose
User Interface	React dashboard / API clients	Collect use-case descriptions and display review results.
Orchestration	AI_AGENT Orchestrator	Call all agents, normalize outputs, aggregate risks and controls, and derive final decision.
Specialized Review	Standards, Safety, Assurance agents	Evaluate different governance dimensions independently.
Model Layer	Ollama + Llama 3.2	Generate role-specific governance assessments locally.

2. Problem Description and Opportunity

The business problem addressed by the project is the growing gap between AI experimentation and institutional governance capacity. Organizations increasingly create AI pilots, AI copilots, automated screening tools, recommendation systems, and internal knowledge agents. However, the review process often remains fragmented: one stakeholder checks privacy, another considers ethics, another reviews technical feasibility, and another asks whether human oversight is sufficient. Without a shared workflow, the organization risks inconsistent approvals, undocumented conditions, and uneven treatment of similar use cases.

This problem is especially important for organizations operating in sensitive domains. AI systems may process personal data, influence access to employment or services, generate public-facing information, or introduce operational dependency on outputs that are probabilistic rather than deterministic. A manual review process can work for a small number of pilots, but it becomes difficult to scale when many teams are experimenting with AI simultaneously. As a result, an institution may either slow innovation through excessive review or approve risky systems without enough evidence.

The opportunity was to build a lightweight governance assistant that makes the first review step more structured. Instead of starting with an empty document, a project sponsor can enter the proposed AI use case, intended users, affected groups, data

sources, deployment context, current controls, and extra context. Specialized agents can then evaluate the proposal from different governance perspectives. The output gives stakeholders an initial map of risks, gaps, required controls, and a decision category.

The project also responds to a documentation problem. AI governance discussions often happen through meetings, emails, or informal notes. Those artifacts are hard to compare across projects. The UNICC AI Agent prototype generates structured JSON-like output and aggregates findings into a consistent summary. That structure supports auditability because a later reviewer can see which categories were considered and which gaps or controls were identified.

The opportunity is not to automate governance judgment away from humans. The opportunity is to raise the quality of human judgment by giving reviewers a consistent starting point. A multi-agent system can surface issues that a single reviewer might overlook, particularly when a proposed system touches legal, ethical, technical, and operational dimensions at once. This framing makes the project suitable for an applied technical project because it combines a practical organizational problem with a concrete software implementation.

3. Importance of the Project to the Organization

The project is important because AI governance has become an operational capability rather than a purely policy-oriented concern. A policy document may state that AI systems should be fair, transparent, accountable, and privacy preserving, but teams still need mechanisms for applying those principles to real use cases. UNICC AI Agent converts governance principles into an interactive evaluation workflow that can be demonstrated, tested, critiqued, and improved.

For an organization such as UNICC, whose mission involves trusted technology services and institutional reliability, the ability to review AI systems before deployment is strategically relevant. The same organization that supports digital transformation must also protect stakeholders from unsafe or uncontrolled AI adoption. A governance assistant helps show how innovation and risk management can coexist. It gives technology teams a practical method to document concerns without stopping all experimentation.

The project also supports institutional integrity. When AI use cases are reviewed inconsistently, stakeholders may believe approvals are arbitrary. A repeatable evaluation system helps reduce that perception. It creates a common language for risk, controls, evidence gaps, and decision outcomes. Even if human committees remain the final

authority, a structured first-pass assessment can improve the quality of committee deliberation.

Another organizational benefit is knowledge retention. Governance reviewers develop expertise over time, but that expertise can remain tacit. By encoding review categories, prompts, output schemas, and aggregation rules into a working system, the organization captures part of that expertise in a form that can be reused. The tool becomes a training artifact as well as a decision-support artifact. New reviewers can learn what types of evidence and controls should be considered for AI deployment.

The repository also uses local model support, which is important for organizations that cannot send sensitive evaluation details to external model providers. Running Llama 3.2 through Ollama demonstrates a privacy-aware development direction. Although local models still require security hardening and quality validation, they can support a governance process where draft use cases, internal controls, and sensitive scenarios remain within the organization's technical boundary.

Table 3-1. Organizational value summary

Value Area	Contribution
Consistency	Creates repeatable evaluation categories and decision outcomes.

Speed	Provides first-pass governance analysis before formal review meetings.
Auditability	Produces structured risks, gaps, controls, and raw/parsed outputs.
Privacy Direction	Supports local model inference instead of default external API calls.
Learning	Acts as a training artifact for AI governance review.

4. Alternate Solutions Evaluated

Several alternative approaches were considered. The first alternative was a fully manual governance checklist. A spreadsheet or document template could capture risk questions, standards references, sign-offs, and required controls. This approach would be low cost and easy to distribute, but it would not provide interactive reasoning, automatic synthesis, or cross-agent aggregation. It would also depend heavily on reviewer discipline, making consistency difficult to maintain across many teams.

The second alternative was a single-agent chatbot. A single LLM prompt could ask one model to evaluate a use case against all governance dimensions. This option would be simpler to build than a multi-agent architecture. However, it would also concentrate all reasoning in one prompt and make it harder to separate legal/standards analysis from safety scoring and assurance review. A single model answer might appear complete while hiding weaknesses in specific domains.

The third alternative was a commercial AI governance platform. Commercial platforms can offer mature features such as user management, model inventory, workflow routing, regulatory updates, and evidence repositories. They may be appropriate for production enterprise deployment. However, they are less suitable for an academic applied prototype because they can obscure how the evaluation process

works internally. They may also introduce cost, vendor dependence, and configuration complexity.

The fourth alternative was a static compliance rules engine. This type of system could classify a use case based on predefined questions and rule thresholds. It would be more deterministic than an LLM-based approach and easier to audit. Its weakness is that governance review often requires qualitative interpretation. For example, the meaning of human oversight depends on context, data sensitivity, affected stakeholders, and deployment authority. A purely static rules engine may not capture these nuances.

The selected approach was a multi-agent LLM-based governance evaluation system. It balances structure and flexibility. The agents are constrained by role-specific prompts and output schemas, but they can still interpret narrative use-case descriptions. The orchestrator provides a deterministic aggregation layer, reducing the risk that the final decision is merely an unstructured model opinion. This design also supports future expansion because additional agents can be added without redesigning the entire system.

5. Solution Evaluation Criteria

The solution was evaluated against criteria that reflect both project feasibility and organizational usefulness. The first criterion was governance coverage. The system needed to address more than general safety; it needed to include standards alignment, safety risk, assurance evidence, accountability, and final decision logic. A narrow classifier would not be sufficient for the applied project because AI governance is multidimensional.

The second criterion was explainability. The output needed to show risks, gaps, controls, and reasoning rather than only a decision label. A governance tool is less useful if stakeholders cannot understand why a use case was approved or rejected. The repository supports explainable outputs through parsed agent responses and summary fields such as cross-agent risks, gaps, and controls.

The third criterion was modularity. The system should be maintainable as policies, standards, and organizational needs evolve. Specialized agents and YAML configuration files support this criterion because roles, prompts, ports, and required output sections can be updated independently. A modular design also helps future teams add specialized reviewers, such as privacy, cybersecurity, procurement, or human rights agents.

The fourth criterion was local deployment feasibility. The prototype should run without depending on external model APIs. This criterion mattered because governance review can involve sensitive institutional details. Local inference through Ollama and Llama 3.2 supports a more privacy-conscious implementation path, even though production deployment would require further security review and performance testing.

The fifth criterion was demonstration readiness. The project needed to be understandable to a sponsor, professor, or reviewer. FastAPI endpoints, Swagger documentation, sample payloads, demo output, and a dashboard-oriented architecture help make the project demonstrable. A system that cannot be shown working would not satisfy the applied project objective, even if the idea were conceptually strong.

Table 5-1. Solution evaluation criteria

Criterion	Reason	Selected Design Response
Governance coverage	AI risks cross legal, safety, and operational domains.	Three specialized agents plus summary aggregation.
Explainability	Reviewers need reasons, gaps, and controls.	Parsed outputs include analysis sections and recommendations.
Modularity	Standards and policies change over time.	YAML configuration and prompt files are separated by agent.
Local feasibility	Sensitive use cases should not require external API disclosure.	Ollama and Llama 3.2 support local inference.

Demonstration readiness	The course project requires a concrete working artifact.	FastAPI endpoints, sample payload, and demo output support review.

6. Selection Rationale

The multi-agent architecture was selected because it most closely matched the real-world structure of AI governance review. In practice, a proposed AI use case is not reviewed by one generic authority. Legal, compliance, risk, data, security, ethics, business, and operational stakeholders each bring a different lens. The selected architecture simulates this distributed review pattern while preserving a single orchestration layer for coordination.

The selected approach also allowed the project to demonstrate technical breadth. The implementation includes Pydantic schemas, FastAPI services, environment-based configuration, YAML agent definitions, prompt files, local LLM calls, output parsing, normalization, aggregation, and a frontend path. This breadth is appropriate for an applied technical project because it shows end-to-end system thinking rather than only prompt writing.

A local open-weight model path was selected because it supports the project's governance message. If the tool is meant to evaluate sensitive AI use cases, it should not require every use case description to be sent to an external service by default. Local deployment does not eliminate all risk, but it aligns better with the principle of data minimization and allows an organization to prototype governance workflows without creating additional third-party exposure.

The orchestrator decision logic was selected to make final decisions more predictable. The repository defines decision categories of Approve, Approve with Conditions, Flag for Review, and Reject. The orchestrator uses agent recommendations to produce a final outcome, with reject and flag outcomes prioritized over simple approval. This design reduces ambiguity and creates a clear governance signal for the reviewer.

The project therefore selected a practical middle path. It is more powerful than a manual checklist, more transparent than a single unstructured chatbot, more educational than a black-box commercial platform, and more flexible than a static rules engine. It is not yet a production governance platform, but it is a defensible applied prototype for demonstrating how multi-agent AI can support institutional AI review.

7. Approach and Methodology

The project methodology followed an applied design-build-evaluate cycle. The first phase defined the governance problem and the desired review workflow. The second phase translated that workflow into system components. The third phase implemented the backend and agent configuration. The fourth phase tested the system with sample use cases. The fifth phase critiqued the result and identified future work.

Requirements were organized around the expected inputs and outputs of an AI governance review. The input schema includes a use case title, description, intended users, affected groups, data sources, deployment context, current controls, and extra context. These fields were sufficient to test the review workflow while keeping the prototype manageable. The schema also encouraged users to provide information that governance reviewers commonly need.

The backend approach used FastAPI because it supports rapid API development, interactive documentation, and straightforward JSON request handling. Each agent service loads its configuration from YAML, loads its system prompt from a prompt file, builds a combined evaluation prompt, calls the local LLM endpoint, parses model output, and returns both raw text and parsed JSON. This pattern makes each agent independently testable.

The orchestrator approach used a central service on port 8080 to call all configured agents. It normalizes outputs, extracts cross-agent risks, gaps, and controls, and derives a final decision. The aggregation method reflects a conservative governance posture: if any agent recommends rejection, the final decision becomes Reject; if any agent flags the use case and no agent rejects it, the final decision becomes Flag for Review; if any agent requires conditions, the final outcome becomes Approve with Conditions.

Testing used a scenario-based method. The repository includes a sample AI Hiring Screening System payload that describes automated ranking of job applicants using resume text, application history, and screening questions. This scenario is appropriate because hiring is a sensitive domain with fairness, transparency, privacy, and accountability concerns. The demo output shows the system producing an Approve with Conditions decision, cross-agent risks, gaps, and controls. This result is consistent with the expected behavior of a governance assistant: it does not ban every high-impact AI use case, but it requires controls before deployment.

Fig. 7-1. Governance evaluation workflow

Step	Workflow Activity	Output
1	Submit AI use case	Structured request payload.
2	Agent review	Standards, safety, and assurance findings.

3	Normalization	Consistent lists of risks, gaps, and controls.
4	Aggregation	Cross-agent summary.
5	Decision	Approve, Approve with Conditions, Flag for Review, or Reject.

8. Project Objectives and Metrics for Project Evaluation

The primary goal was to create a working multi-agent AI governance evaluation prototype that can assess proposed AI use cases and produce structured, explainable recommendations. The project objectives were defined at a level appropriate for a course project: functional completeness, governance coverage, explainability, local deployment feasibility, and demonstration readiness.

Objective one was to implement a working backend capable of receiving an AI use-case description and returning agent-specific evaluations. This objective was met through the FastAPI agent service and the shared EvaluateRequest schema. A reviewer can submit a JSON payload and receive raw and parsed agent output. This demonstrates that the governance evaluation process is operational rather than purely conceptual.

Objective two was to implement multi-agent specialization. This objective was met through three configured agents: Standards Compliance, Safety Dimensions, and Assurance & Accountability. Each agent has its own role, prompt file, required sections, and endpoint configuration. This design demonstrates separation of concerns and supports later extension.

Objective three was to implement aggregation and final decision logic. This objective was met through the orchestrator service, which calls configured agents, normalizes result structures, summarizes risks, gaps, and controls, and returns a final decision. The

available decision outcomes are meaningful for governance workflow because they correspond to approve, conditional approve, escalate, or reject actions.

Objective four was to demonstrate the solution with a realistic test scenario. This objective was met through the sample AI Hiring Screening System payload and demo output. The output included high-priority risks such as harmfulness, privacy, and accountability; gaps related to human oversight, security, diversity, transparency, and documentation; and controls such as bias testing, audit logging, secure data handling, human oversight, and monitoring.

Table 8-1. Project objectives and evaluation metrics

Objective	Metric	Result
Working evaluation API	A use-case payload can be submitted and evaluated.	Met through FastAPI /evaluate endpoints.
Multi-agent review	At least three distinct governance agents are configured.	Met: standards, safety, assurance.
Final decision logic	Orchestrator returns consolidated decision.	Met through /run endpoint and decision aggregation.
Explainable output	Output includes risks, gaps, controls, and reasoning.	Met in demo output and parsed agent responses.

Local model path	Prototype can use local Llama 3.2 through Ollama.	Met according to repository setup and LLM client.

9. Risk Analysis

The project included several technical, governance, and adoption risks. The first technical risk was model output inconsistency. LLMs can return malformed JSON, omit required fields, or produce reasoning that sounds plausible but is not grounded in current standards. The system mitigates part of this risk through parsing logic, normalization functions, required output sections, and an orchestrator that structures results. However, additional validation would be needed for production use.

The second technical risk was local inference performance. Running Llama 3.2 through Ollama can require meaningful RAM and compute resources. A laptop or small server may experience latency, especially when multiple agents are called sequentially. The project mitigated this risk by using a local model appropriate for prototype use, but a production environment would need hardware sizing, concurrency planning, caching, and timeout management.

The third risk was governance overconfidence. A polished output could lead users to treat the tool as an authority rather than a decision-support system. The project mitigates this conceptually by framing final decisions as recommendations and by emphasizing human oversight. Future versions should make this boundary explicit in the user interface and require human sign-off before any deployment decision is recorded.

The fourth risk was standards drift. AI regulation and best practices are changing quickly. A governance assistant that relies on static prompt text may become outdated. The project mitigates this by using modular prompt files and configuration, but future work should include a maintained knowledge base, source citations, versioning, and review dates for all standards references.

The fifth risk was incomplete organizational integration. A prototype can generate useful assessments, but institutional value depends on workflow adoption. If the tool is not integrated into intake, approval, monitoring, and exception processes, its output may remain an academic artifact. The project mitigates this through clear decision categories and structured outputs, but production adoption would require governance ownership, change management, and integration with project management or ticketing systems.

Table 9-1. Risk register

Risk	Likelihood	Impact	Mitigation
Malformed or inconsistent LLM output	Medium	High	Use parsing, normalization, schema validation, and future retry logic.
Outdated standards references	High	High	Version prompts and connect future retrieval

			to verified standards sources.
Overreliance on tool recommendation	Medium	High	Require human approval and show decision-support disclaimers.
Local model latency	Medium	Medium	Optimize model size, hardware, timeouts, and concurrency.
Sensitive input exposure	Low to Medium	High	Run locally, add access controls, logging policy, and secure storage.

10. Issues Encountered

One issue encountered was the challenge of converting broad AI governance principles into a concise but useful set of agent roles. Governance frameworks often contain overlapping concepts such as transparency, explainability, accountability, risk management, human oversight, and fairness. The project addressed this issue by assigning related concerns to separate agents while allowing the orchestrator to consolidate overlapping findings.

A second issue was output normalization. LLM responses can represent lists, dictionaries, and recommendation text in inconsistent ways. The code addresses this issue by normalizing safety dimensions, high-priority risks, mitigations, key gaps, required controls, evidence gaps, assurance controls, and accountability structures. This was an important implementation step because aggregation depends on predictable structures.

A third issue was determining how conservative the final decision logic should be. If the system approved too easily, it would fail as a governance assistant. If it rejected too often, it would discourage innovation and reduce adoption. The selected logic is conservative but practical: rejection takes precedence, flagging triggers escalation, conditional approval is the default when any agent requires controls, and full approval requires agreement across agents.

A fourth issue was distinguishing prototype success from production readiness. The prototype can demonstrate the workflow, but it does not include all features expected in an enterprise system. For example, it lacks persistent user accounts, role-based access control, a database of reviews, evidence attachments, formal audit trails for orchestrator runs, and legal review integration. This issue is handled in the report by clearly separating completed scope from future work.

A fifth issue was the inherent uncertainty of AI governance evaluation. The system can identify likely gaps and controls, but it cannot verify whether an organization has actually implemented those controls. For example, it may recommend bias testing or audit logging, but a human reviewer must inspect evidence. This limitation reinforced the conclusion that the tool should support governance review, not replace it.

11. Detailed Project Chronology and Critique

The project began with problem framing. The initial question was how to turn AI safety and governance theory into a working tool that could be demonstrated. The project scope was narrowed to an intake-and-evaluation workflow because this stage is where many organizations struggle. A pre-deployment review tool is also feasible for an applied project because it can be tested with sample use cases without requiring a live production AI system.

The next stage was architecture selection. A multi-agent pattern was selected because governance review benefits from specialized perspectives. The architecture separated standards compliance, safety dimensions, and assurance accountability. This separation helped define prompts, outputs, and responsibilities. It also created a clear story for the final report: each agent acts like a governance reviewer, while the orchestrator acts like a committee chair.

The implementation stage focused on backend services and configuration. The agent app loads YAML configuration, retrieves a prompt file, constructs a prompt from the submitted use case, calls the local model, attempts JSON parsing, and returns results. The LLM client sends a non-streaming request to Ollama with a controlled temperature setting. The schema file defines the EvaluateRequest fields that structure user input.

The orchestration stage implemented the system-level behavior. The orchestrator reads its configuration, calls each agent endpoint, normalizes outputs, builds a summary, and derives a final decision. This was a critical step because separate agent outputs are not enough for a governance workflow. Decision makers need a consolidated answer that highlights the most important cross-agent risks, gaps, and controls.

The demonstration stage used the AI Hiring Screening System scenario. This was a strong test case because hiring is a sensitive area where fairness, privacy, transparency, and accountability risks are easy to understand. The result of Approve with Conditions showed that the system can produce a nuanced decision. It identified gaps and controls without treating the use case as automatically unacceptable.

In critique, the project could have improved in several ways. First, a formal test set with expected outcomes would have made evaluation stronger. Second, a stronger frontend demonstration would have made the user experience clearer. Third, a persistent review database would have supported longitudinal auditability. Fourth, automated JSON schema validation and retry logic would have increased robustness. Finally, direct citations to standards in agent outputs would have strengthened trust.

Fig. 11-1. Baseline project chronology

Phase	Main Work	Critique

Problem framing	Defined need for AI governance intake and review.	Could have included more stakeholder interviews.
Architecture	Selected multi-agent pattern and orchestrator.	Strong conceptual fit, but adds operational complexity.
Implementation	Built FastAPI agents, schemas, LLM client, and orchestrator.	Needs stronger automated tests.
Demonstration	Ran hiring-screening scenario and generated decision output.	More scenarios would improve confidence.
Reflection	Identified limitations and future development plan.	Production readiness remains future scope.

12. Lessons Learned

The first lesson learned is that AI governance cannot be reduced to one checklist question. A system may be technically impressive while still lacking transparency, privacy safeguards, accountability structure, or sufficient evidence. The multi-agent design made this visible because each reviewer role surfaced different concerns. This lesson is important for any organization deploying AI: governance needs multiple lenses.

The second lesson is that prompt engineering and software engineering must work together. A prompt can request structured output, but production-like use requires schemas, parsing, normalization, timeouts, configuration, and error handling. The project showed that an AI governance assistant is not simply a prompt; it is a system that must manage uncertain model behavior and turn it into useful workflow output.

The third lesson is that local model deployment has strategic value but introduces operational tradeoffs. Running a model locally supports privacy and reduces external dependence, but it creates performance, installation, hardware, and maintenance challenges. Organizations considering local SLM or LLM deployment should evaluate both governance benefits and infrastructure burden.

The fourth lesson is that explainability must be designed into outputs. A decision label alone is insufficient. Reviewers need to see which standards were triggered, which safety dimensions scored highly, which evidence gaps remain, which controls are

required, and what residual risk remains. The project design reinforced that governance communication is as important as governance classification.

The fifth lesson is that applied project success depends on scope control. A complete enterprise AI governance platform would require months or years of development. The project succeeded by selecting a bounded scope: intake, specialized evaluation, aggregation, and demonstration. That scope was large enough to show value and small enough to implement as a course project.

13. Conclusions and Summary

The UNICC AI Agent project demonstrates that a multi-agent AI system can support structured evaluation of proposed AI use cases. The project produced a working reference architecture with specialized agents, local model inference, FastAPI endpoints, orchestrated aggregation, and decision categories aligned with governance workflow. It is a practical proof of value for institutional AI review.

The strongest contribution of the project is its translation of governance principles into system behavior. Instead of merely stating that AI systems should be fair, transparent, accountable, and safe, the prototype asks for use-case context, evaluates the proposal through distinct roles, and returns specific gaps and controls. This creates a bridge between policy language and operational decision making.

The project also demonstrates the value of modularity. The agent configuration files, prompt files, schema definitions, and orchestrator logic can be updated independently. This matters because AI governance is not stable. Regulations, standards, organizational policies, and best practices will continue to evolve. A modular design is more adaptable than a fixed monolithic prompt or a static checklist.

At the same time, the project has clear limitations. It should not be used as a final approval authority. It does not verify evidence, cannot guarantee legal accuracy, and may produce inconsistent reasoning if the underlying model changes. Its output should

be reviewed by qualified humans. The tool is therefore best positioned as a first-pass assessment and documentation aid.

Overall, the project met its applied objective. It created a functioning prototype that addresses a real governance need, demonstrates technical implementation, supports explainable review, and provides a credible foundation for future development. The recommended next step is to convert the prototype into a workflow system with persistent storage, user roles, evidence attachments, standards citations, testing dashboards, and sponsor-approved acceptance criteria.

14. Limitations, Recommendations, and Scope for Future Work

The first limitation is that the system relies on model-generated reasoning. Even with a low temperature setting and structured prompts, the model can hallucinate, omit relevant standards, or misinterpret a use case. Future work should add retrieval-augmented generation from verified standards documents and require citations for all regulatory claims.

The second limitation is that the current prototype does not include persistent workflow management. A production governance tool should store each submitted use case, version the review, attach evidence, record reviewer comments, and track whether required conditions were satisfied. Without persistence, the prototype is best suited for demonstration and early analysis rather than institutional records management.

The third limitation is limited security hardening. A production version should include authentication, authorization, role-based access control, input filtering, logging, encrypted storage, secure deployment configuration, and operational monitoring. This is especially important because the tool may process sensitive descriptions of proposed AI systems, data sources, controls, and organizational weaknesses.

The fourth limitation is that the project does not include a large empirical evaluation. It uses sample scenarios and demo output, but a stronger evaluation would include a benchmark set of use cases across hiring, finance, health, customer support, education,

public sector services, and internal productivity. Each case could be reviewed by human experts to compare model recommendations with expert judgments.

Future work should focus on six improvements: standards retrieval, evidence management, human approval workflow, test-set evaluation, user interface refinement, and deployment hardening. These additions would move the project from a course-ready prototype toward a more mature governance assistant. The long-term vision is a system that helps organizations maintain an AI inventory, review new use cases, monitor controls, and generate audit-ready governance records.

15. Literature Survey

The literature and standards context for this project shows that AI governance has converged around several recurring themes: risk management, transparency, accountability, privacy, fairness, human oversight, monitoring, and documentation. These themes appear in government frameworks, international recommendations, and technical standards. The project drew on this convergence by designing agents that evaluate standards compliance, safety dimensions, and assurance accountability.

The NIST AI Risk Management Framework provides one of the most practical reference models for the project. NIST describes the AI RMF as a resource to help organizations that design, develop, deploy, or use AI systems manage AI risks and promote trustworthy and responsible AI. The framework is organized around functions such as Govern, Map, Measure, and Manage. The UNICC AI Agent prototype aligns with this logic by requiring contextual information, identifying risks and gaps, and recommending controls.

The EU AI Act provides a regulatory example of risk-based AI governance. The repository references the EU AI Act as one of the standards considered by the Standards Compliance Agent. The Act is relevant because it emphasizes risk categories, requirements for high-risk AI systems, transparency, human oversight, and obligations

for different actors. Even when an organization is not directly subject to the Act, its structure provides a useful model for classifying and controlling AI risks.

UNESCO's Recommendation on the Ethics of Artificial Intelligence provides a broader human-centered frame. UNESCO emphasizes human rights, dignity, transparency, fairness, and human oversight. These principles are reflected in the project's safety dimensions and assurance concerns. They also justify the project's emphasis on affected groups and deployment context, because ethical review requires attention to who may be affected by AI decisions.

ISO/IEC 23894:2023 provides guidance on AI risk management for organizations that develop, produce, deploy, or use AI systems. Its relevance to the project is that governance is not merely a one-time ethical statement; it is a risk management process that should be integrated into organizational activities. The UNICC AI Agent prototype supports that integration by creating a repeatable intake and review workflow.

The literature also suggests that governance tools must be auditable and accountable. Responsible AI programs require evidence, not only assertions. A system that recommends controls should also track whether those controls exist and whether they are effective. This insight informed the future-work recommendation for evidence attachments, review records, and human sign-off. The prototype begins the process by

producing structured outputs, but a mature version must connect those outputs to actual proof.

Finally, the rise of multi-agent and orchestrated LLM systems influenced the architecture. Multi-agent systems can decompose complex tasks into specialized reasoning roles. However, they also require coordination, error handling, and aggregation logic. The UNICC AI Agent project demonstrates this principle in a governance setting: each agent analyzes a different dimension, while the orchestrator produces a consolidated decision. The design therefore reflects both responsible AI literature and contemporary AI system architecture practice.

16. Situational Analysis

The situational analysis considers the environment in which the project would be used. The organization faces rising demand for AI experimentation, increasing regulatory scrutiny, and growing concern about institutional trust. Teams want to deploy AI quickly, but leaders need a way to ensure that high-impact use cases are reviewed before they create harm. This tension creates the need for a governance assistant that is fast enough for innovation and structured enough for accountability.

Internal strengths include access to technical expertise, a clear educational environment, and a project sponsor familiar with applied technology evaluation. The repository demonstrates implementation capability through FastAPI services, agent configuration, local LLM integration, and orchestration logic. These strengths make it realistic to continue improving the prototype.

Internal weaknesses include the prototype nature of the current system, limited testing, and missing production workflow features. The tool does not yet include authentication, persistent records, reviewer permissions, standards retrieval, evidence storage, or automated report generation. These weaknesses do not invalidate the project, but they define the boundary between course prototype and deployable governance platform.

External opportunities include the rapid growth of institutional demand for AI governance tools. Many organizations need practical methods for AI inventory, review, control mapping, and audit preparation. A modular local-governance assistant could be valuable for organizations that want to experiment without sending sensitive use-case data to external APIs. The project could also support training, simulation, and classroom demonstrations.

External threats include regulatory change, model reliability concerns, competing commercial platforms, and stakeholder resistance. Regulations may change faster than prompts are updated. Users may distrust model-generated governance advice. Commercial platforms may offer more complete features. Teams may resist governance intake if they perceive it as slowing innovation. These threats reinforce the recommendation that the tool should be positioned as an aid to better review, not as a bureaucratic barrier.

The situational conclusion is that the project is well matched to an early-stage institutional need. It should not be sold as a finished enterprise compliance platform, but it is a strong applied prototype. Its value lies in showing how AI governance can be operationalized through structured intake, specialized evaluation, aggregation, and explainable decision categories.

Appendices

Appendix A. Project Acceptance

See the following page.

Sponsor's Project Acceptance Document – PART 1

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. It is therefore important that you describe the goal, objectives, and related metrics in the appropriate section below. The "PLAN" section is to be prepared at the beginning of the project and the "RESULTS" and "ACCEPTANCE" sections after your project has been completed. Your sponsor should provide input and sign where indicated. The signed document will also be a required section in your Project Final Report. This document is a template whose sections may be expanded as necessary.

Project Title: UNICC AI Safety Lab

Student Name: Yizhuo Shi

Sponsoring Organization: NYU SPS, United Nations International Computing Centre UNICC

Project Sponsor Name and Title: Dr. Andrés Fortino Associate Clinical Professor

Project Sponsor Contact Information (email and phone): afortino@nyu.edu

PROJECT PLAN

At project start, show the project goal; the project objectives and related metrics to be used to show successful project completion. Sponsor should sign to indicate agreement.

Project Goal: The goal of this project is to develop a standalone on premises AI safety lab for the UNICC using a Mixture of Experts architecture arbitrated by a fine tuned Small Language Model to evaluate AI risks without compromising data sovereignty.

Project Objective #1: Orchestration Logic & Disagreement Surfacing

Refactor and integrate the "Council of Experts" orchestration layer to enable disagreement surfacing and synthesis between specialized safety modules by April 14, 2026.

Project Objective #2: Benchmark Alignment & Accuracy

Fine-tune the Llama-3-8B model on curated Governance, Risk, and Compliance (GRC) datasets to achieve a minimum of 90% alignment with established UNICC safety benchmarks by April 14, 2026.

Project Objective #3: Performance Optimization (Latency)

Implement 4-bit quantization on the fine-tuned Small Language Model (SLM) to reduce average inference latency to under 2 seconds per safety assessment report by April 14, 2026.

Project Objective #4: GitHub Link Ingestion Capability

Develop a capability for the "Council of Experts" to fetch, parse, and assess data directly via public GitHub repository links by April 20, 2026.

Project Objective #5: Technical Handover & Documentation

Compile a comprehensive technical handover package including optimized model weights and a finalized GitHub repository by April 25, 2026.

I agree with the above planned project goal, project objectives, and related metrics.

Andres Fortino

Project Sponsor Signature **Date:**

Printed Name: **Dr. Andrés Fortino**

Please print in English

Sponsor's Project Acceptance Document -Part 2

PROJECT RESULTS

STUDENT: Include below the project goal and ALL the objectives in your project, which you detailed in Project Plan section (07-Part1). You may add additional objectives, or add space, as needed.

Planned Start Date: Feb 9, 2026 **Planned End Date:** April 25, 2026
Actual Start Date: Feb 9, 2026 **Actual End Date:** April 25, 2026

If actuals differ from planned dates, the revised dates (Actual) are accepted by the sponsor if initialed here: **Sponsor Initials** _____

Project Goal Was the project goal achieved as planned? ☑Yes ☐No, Reason missed: _____
If NO, please explain why this is an acceptable deviation. _____ **Sponsor Initials** AF

Project Objective #1: <as shown above in Plan section (Part1)>
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#1** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #2: <as shown above in Plan section (Part1)>
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#2** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #3: <as shown above in Plan section (Part1)>
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#3** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #1: Orchestration Logic & Disagreement Surfacing

Refactor and integrate the "Council of Experts" orchestration layer to enable disagreement surfacing and synthesis between specialized safety modules by April 14, 2026.

Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#1** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #2: Benchmark Alignment & Accuracy

Fine-tune the Llama-3-8B model on curated Governance, Risk, and Compliance (GRC) datasets to achieve a minimum of 90% alignment with established UNICC safety benchmarks by April 14, 2026.

Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#2** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #3: Performance Optimization (Latency)

Implement 4-bit quantization on the fine-tuned Small Language Model (SLM) to reduce average inference latency to under 2 seconds per safety assessment report by April 14, 2026.

Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#3** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #4: GitHub Link Ingestion Capability

Develop a capability for the "Council of Experts" to fetch, parse, and assess data directly via public GitHub repository links by April 20, 2026.

Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#4** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #5: Technical Handover & Documentation

Compile a comprehensive technical handover package including optimized model weights and a finalized GitHub repository by April 25, 2026.

Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#5** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

PROJECT ACCEPTANCE

Sponsor: Please show your evaluation of student's work, add comments, sign below

SPONSOR: Select One

Sponsor's evaluation of student's performance on this project:
☑ **A** ☐ **A-** ☐ **B+** ☐ **B** ☐ **B-** ☐ **C+** ☐ **C** ☐ **C-** ☐ **F**

Sponsor's Comments: _____

_____

SPONSOR: Select One

☑ Project was completed satisfactorily and is hereby accepted.

☐ Project was completed satisfactorily but did not meet all objectives, as shown above. Nevertheless, the Project is accepted.

☐ Project was NOT completed satisfactorily and is NOT accepted.
Please explain why not accepted in Sponsor's Comments section, above.

Andres Fortino

Project Sponsor Signature **Date:**

Student Signature **Date:**

All signatures in English. Sign by hand or use e-signature. Do not type or use handwriting font.

Appendix B. Project Plan, Baseline and Actual WBS

The baseline project plan organized the work into problem definition, architecture design, implementation, testing, documentation, and final reporting. The actual work followed the same general sequence, but additional effort was spent on output normalization and decision aggregation because these elements were more important than initially expected.

Table B-1. Baseline and actual WBS

WBS Item	Baseline Plan	Actual Result	Deviation
1. Requirements	Define governance review inputs and outputs.	Completed with EvaluateRequest schema and decision categories.	Minor; scope remained stable.
2. Architecture	Design multi-agent evaluation flow.	Completed with three agents and orchestrator.	No major deviation.
3. Backend	Implement API endpoints.	Completed with agent /evaluate and orchestrator /run endpoints.	Additional normalization added.

4. Model integration	Use local LLM through Ollama.	Completed through LLM client using /api/generate.	No major deviation.
5. Testing	Run sample governance scenarios.	Completed with hiring-screening scenario and demo output.	More scenarios deferred.
6. Documentation	Prepare README and final report.	Completed draft final report and project references.	Sponsor acceptance remains pending insertion.

Appendix C. Supporting Project Evidence

The supporting evidence for this final report includes the public GitHub repository for UNICC-AI-Agent, the sample payload, the demo output, and the described architecture. The repository files reviewed for this report include README.md, agents/app.py, agents/llm.py, agents/schemas.py, orchestrator/app.py, orchestrator/aggregate.py, the YAML configuration files, scripts/run_all.sh, examples/sample_payload.json, and demo_output.json.

Table C-1. Repository evidence map

Artifact	Evidence Provided

README.md	Overview, key features, architecture, governance dimensions, setup, testing, and notes.
agents/app.py	Agent service, prompt construction, JSON parsing, normalization, health, and evaluate endpoint.
agents/llm.py	Ollama local generation client and low-temperature model option.
agents/schemas.py	Pydantic request and response models.
orchestrator/app.py	Agent calling, result normalization, summary construction, and final decision logic.
configs/*.yaml	Agent roles, models, endpoints, required output sections, and orchestrator routing.
demo_output.json	Demonstration of final decision, risks, gaps, controls, and parsed agent output.

Appendix D. References and Bibliography

European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union.

hisiwen. (2026). UNICC-AI-Agent [GitHub repository]. GitHub. https://github.com/hisiwen/UNICC-AI-Agent

International Organization for Standardization. (2023). ISO/IEC 23894:2023, Information technology - Artificial intelligence - Guidance on risk management.

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0) (NIST AI 100-1). U.S. Department of Commerce.

UNESCO. (2022). Recommendation on the ethics of artificial intelligence. United Nations Educational, Scientific and Cultural Organization.

White House Office of Science and Technology Policy. (2022). Blueprint for an AI Bill of Rights: Making automated systems work for the American people.

Appendix E. Supplemental Implementation Narrative

Governance intake discipline

A useful AI governance process begins with disciplined intake. The project schema forces the submitter to describe purpose, users, affected groups, data sources, deployment setting, and current controls. These categories matter because many AI risks are invisible if the review begins only with a technical model description. A ranking model used for entertainment recommendations creates a different risk profile than a

ranking model used for hiring or benefits decisions. The intake stage therefore functions as a risk discovery mechanism.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Standards interpretation

The Standards Compliance Agent represents the standards interpretation function. Its role is not to act as a lawyer, but to surface likely obligations and governance expectations. This is valuable because teams often do not know which standards or regulations may be relevant to a proposed AI use case. By producing triggered standards, key gaps, and required controls, the agent gives the human reviewer a starting point for deeper analysis.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Safety dimension scoring

The Safety Dimensions Agent makes qualitative risk visible through named dimensions. Harmfulness, bias and fairness, transparency, deception, privacy, and accountability are practical categories because they are understandable to non-specialists while still mapping to serious governance concerns. The scoring approach helps prioritize review attention. High scores should trigger additional evidence requests, mitigation planning, and potentially escalation.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept

but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Assurance and accountability

The Assurance & Accountability Agent addresses the question that often separates policy from implementation: what evidence proves the system is ready? A use case may claim that humans are in the loop, but the governance reviewer must know who owns the decision, how exceptions are handled, what monitoring exists, what audit logs are retained, and how incidents are escalated. This agent encourages concrete operational proof.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence

review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Decision categories

The four decision categories create a clear governance vocabulary. Approve means the use case appears ready within the available evidence. Approve with Conditions means deployment may proceed only after specified controls are implemented. Flag for Review means the case needs escalation or expert review. Reject means the proposed design is not acceptable in its current form. These categories are simple but powerful because they translate analysis into action.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Human-in-the-loop boundary

The project reinforces the boundary between automated analysis and human authority. The tool can summarize risks and suggest controls, but it cannot determine organizational risk appetite, legal obligations, stakeholder priorities, or moral tradeoffs with finality. A production version should require named human reviewers to accept or modify recommendations. This would preserve accountability and prevent governance automation from becoming governance theater.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide

better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Auditability and evidence

Auditability is essential for responsible AI governance. A future version should store every input, agent output, model version, prompt version, reviewer decision, evidence attachment, and control status. This would allow an organization to explain why a use case was approved and what conditions were required. The current prototype begins this journey by producing structured outputs, but durable audit records remain future scope.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions

more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Scalability of review

The project addresses scalability by making the first-pass review repeatable. A small governance team cannot manually research every proposed AI use case from scratch. A structured assistant can handle routine synthesis, allowing humans to focus on judgment, evidence, escalation, and stakeholder engagement. This does not reduce the importance of human review; it reallocates human effort toward higher-value decisions.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Production hardening

Before production deployment, the system should be hardened. Required work includes authentication, authorization, encrypted storage, secure secret management, containerized deployment, rate limiting, logging policy, model monitoring, test automation, and disaster recovery planning. Because the tool itself may process sensitive governance information, it should meet a high security standard.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.

Future strategic value

The strategic value of the project lies in its extensibility. Additional agents could evaluate privacy impact, cybersecurity posture, procurement risk, human rights concerns, data quality, model monitoring, or environmental impact. The same orchestrator pattern could become a broader governance platform. The prototype therefore functions as a foundation for future institutional AI review capability.

In practical deployment, this topic would require clear ownership, documented procedures, and measurable acceptance criteria. The project demonstrates the concept but should be expanded through stakeholder interviews, test scenarios, and evidence review. The most important design principle is that governance outputs must be useful to decision makers, not merely impressive as model-generated text.

The lesson for future development is that responsible AI tools must be evaluated by their effect on organizational behavior. A successful tool should cause teams to provide better context, identify risks earlier, implement controls sooner, and document decisions more consistently. Those behavioral outcomes are more important than the elegance of the software architecture alone.